Theoretical probability

1 Amy picks a whole number between 4 and 8 inclusive.

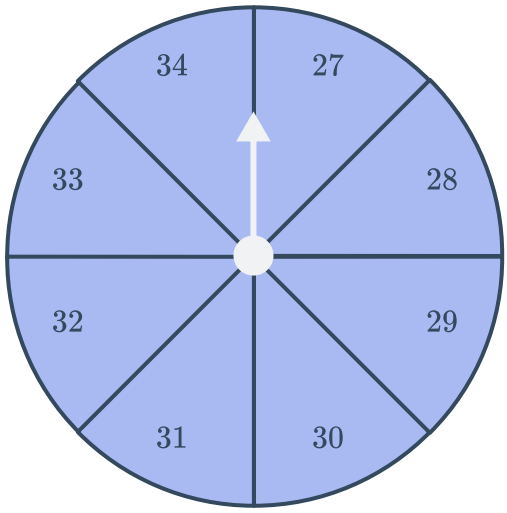
- **a** List the sample space.
- **b** Find the probability that Amy picked the number 5.
- **c** Find the probability that Amy picked the number 1.
- **d** Find the probability that Amy picked an even number.

2 A standard six-sided die is rolled once.

- **a** List the sample space.
- **b** Find the probability of rolling a 2.
- **c** Find the probability of rolling a 6.

3 Charlene spins the spinner shown:

- **a** List the sample space.
- **b** Find the probability of spinning a number greater than 29.
- **c** Find the probability of spinning a number in the twenties.
- **d** Find the probability of spinning a 27 or a 29.



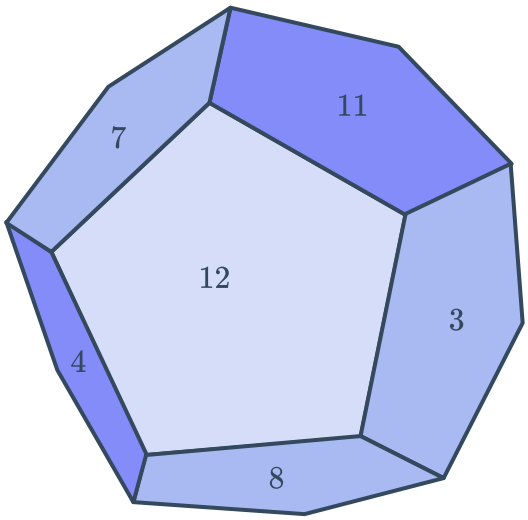
4 A number is randomly selected from the following list:

$$\{1, 3, 3, 6, 6, 6, 8, 8, 8, 8, 10, 10, 10, 10, 10\}$$

- **a** Find the probability of selecting a 1.
- **b** Find the probability of selecting a 3.
- **c** Find the probability of selecting an 8.
- **d** Which number is most likely to be selected?

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5 A 12-sided die has faces with the numbers 1 through 12 as shown:



a Find the probability of rolling an odd number.

b Find the probability of rolling a multiple of 4.

6 A year 4 class are sitting at their desks in the order shown below. A student is chosen at random from the class.

	Column 1	Column 2	Column 3	Column 4	Column 5
Row 1	Laura	Kenneth	Sophia	Hermione	Valentina
Row 2	Oliver		Maria	Harry	Roald
Row 3	Gwen	Jenny		Fred	Mario
Row 4		Aaron	Luke	Elizabeth	Derek

a How many possible outcomes are there?

b Find the probability that a student in Column 4 is picked.

c Find the probability that a student in the Row 2 is picked.

d Find the probability that a student in Year 4 is picked.

7 A bag contains 17 yellow marbles, 10 grey marbles and 14 orange marbles.
If one marble is picked at random from the bag, find the probability that it is:

a Yellow

b Grey

c Orange

d Yellow or orange

e Grey or orange

f Yellow, grey or orange

8 A book has pages numbered from 1 to 100. If the book is opened to a random page, find the probability that the page number:

a Is a multiple of 9.

b Has the digit $'6'$ in the page number.

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9 A cube has six faces, each face is painted a certain colour. If the cube is rolled, find the number of faces that should be painted blue so that the probability of blue appearing on the uppermost face is:

a $\frac{1}{2}$ **b** $\frac{1}{3}$ **c** $\frac{1}{6}$ **d** 1

10 Each 26 letters of the alphabet are written on separate pieces of paper and placed in a bag. If one letter is to be picked out of the bag at random find the probability of picking a:

a J

b K, Y or R

c Letter in the word PROBABILITY

d M, K, D, O, H or B

e Letter in the word WORKBOOK

11 A bag contains 86 marbles, some of them are black and some are white. If the probability of selecting a black marble is $\frac{33}{43}$, find:

a The number of black marbles.

b The number of white marbles.

12 Charlie enters a raffle in which 300 tickets are sold. Find the probability of him winning a prize if he purchases:

a 1 ticket **b** 2 tickets **c** 3 tickets **d** n tickets

13 A circular spinner is divided into three unequal parts. The green sector takes up an angle of 250 degrees at the centre. The red sector takes up an angle of 60 degrees at the centre and the blue sector takes up the remainder of the spinner. Find the probability that the spinner will land on blue.

14 A six-sided die is rolled. Find:

a $P\left(\text{odd number}\right)$

b $P\left(\text{number greater than } 1\right)$

c $P\left(\text{number divisible by } 2\right)$

d $P\left(\text{number less than } 1\right)$

e $P\left(\text{neither } 2 \text{ nor } 5\right)$

15 A marble is randomly drawn from a bag which contains 6 red marbles, 7green marbles and 3 blue marbles. Find:

a $P\left(\text{red}\right) + P\left(\text{green}\right) + P\left(\text{blue}\right)$

b $P\left(\text{red or green}\right)$

c $P\left(\text{red or blue}\right)$

d $P\left(\text{green or blue}\right)$

16 The sample space of an event is listed as $S = \{\text{short}, \text{average}, \text{tall}\}$. If $P\left(\text{average}\right) = 0.5$ and $P\left(\text{short}\right) = 0.3$, find $P\left(\text{tall}\right)$.

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Complementary events

17 For each of the following, state whether the two events are complementary:

a Event 1: Selecting a positive number.
Event 2: Selecting a negative number.

b Event 1: Drawing a red card from a standard deck of cards (no jokers).
Event 2: Drawing a black card from a standard deck of cards (no jokers).

c Event 1: Drawing a club from a standard deck of cards (no jokers).
Event 2: Drawing a spade from a standard deck of cards (no jokers).

d Event 1: Rolling a number greater than 3 on a die.
Event 2: Rolling a number less than 3 on a die.

18 Find each of the following probabilities, find the probablity that the complementary event will occur:

a $\frac{4}{5}$

b 0.64

19 A biased coin is flipped, with heads and tails as possible outcomes. Calculate $P\left(\text{heads}\right)$ if $P\left(\text{tails}\right) = 0.56$.

20 A bag contains 34 red marbles and 35 blue marbles. If picking a marble at random, find:

a $P\left(\text{red}\right)$

b $P\left(\text{not red}\right)$

21 A bag contains 50 black marbles, 37 orange marbles, 29 green marbles and 23 pink marbles. If a marble is selected at random, find the following probabilities:

a $P\left(\text{orange}\right)$

b $P\left(\text{orange or pink}\right)$

c $P\left(\text{not orange}\right)$

d $P\left(\text{neither orange nor pink}\right)$

22 A number between 1 and 100 inclusive is randomly picked.

a State the complement of picking a number greater than 61.

b Find the probability that the number picked is greater than 61.

23 A regular die is rolled. Find the probability of:

a Not rolling a 4.

b Not rolling a 1 or 5.

c Not rolling an even number.

d Not rolling an 8.

e Not rolling a 1, 2, 3, 4, 5, or 6.

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24 The 26 letters of the alphabet are written on pieces of paper and placed in a bag. If one letter is picked out of the bag at random, find the probability of:

a Not selecting a B.

b Not selecting a K, R or T

c Selecting a letter that is not in the word PROBABILITY

d Not selecting a T, L, Q, A, K or Z

e Selecting a letter that is not in the word WORKBOOK

25 A card is drawn at random from a standard deck. Find the probability that the card is:

a A diamond **b** A spade **c** Not a heart

26 From a normal deck of cards, find the probability of:

a Selecting a five

b Selecting a nine

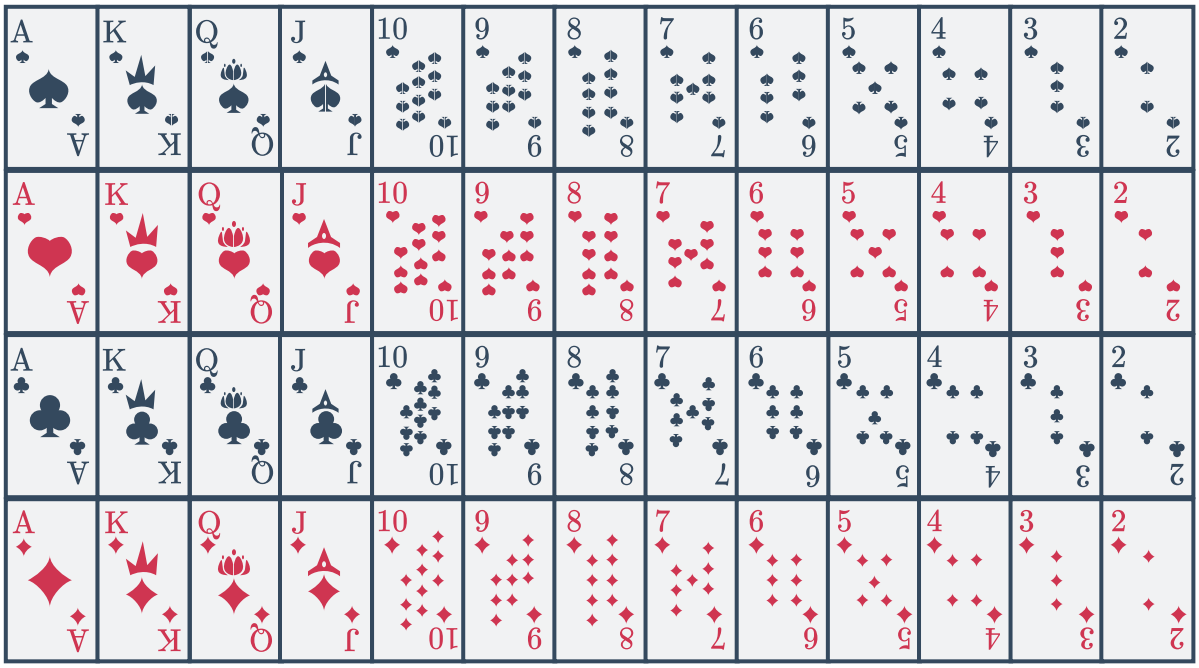
c Not selecting a two

d Selecting a black card

e Not selecting a black card

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27 A card is selected from a standard deck of cards:



Find the probability of:

a Selecting a face card.

b Selecting a black nine.

c Selecting an odd-numbered black card, not counting ace as a numbered card.

d Selecting a red nine.

e Not selecting a red three.

f Not selecting a queen of clubs

g Selecting a ten, jack, queen, king or ace

h Selecting the king of diamonds

i Not selecting a red ten or black jack

28 The number of movie, concert and musical tickets that are offered as a prize in a raffle are in the ratio $8 : 19 : 3$.

a Calculate the probability that the winner will be given a concert ticket.

b The winner doesn't want to see a musical. Calculate the probability that they get a ticket they want.

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